

WorldRoll: From Privileged Camera Trajectories to Pose-Free World Rollouts

Anonymous submission

Abstract

Precise camera control in video world models is commonly obtained from numeric poses, trajectories, or action tensors, although users often have only an image and a natural-language intent. We present WorldRoll, a data-centric framework that transfers privileged camera supervision into pose-free image-to-video (I2V) world rollouts. Here, pose-free means that the fine-tuned student receives no numeric pose, explicit trajectory, or action tensor during training or inference; camera intent is conveyed only through language. WorldRoll constructs complementary supervision from pose-aware real video, dynamics-rich real video, and videos generated by a camera-controllable teacher from real start frames. A shared camera language converts heterogeneous evidence into a structured annotation and a deterministic canonical prompt, while a VLM quality gate rejects clips with fatal validity failures or weak visual, temporal, geometric, grounding, or control quality. The resulting first-frame-prompt-video triples fine-tune two compact open-weight I2V backbones, LTX-Video-2B and CogVideoX-Fun-2B, with LoRA under their native objectives. At inference, a prompt rewriter canonicalizes free-form instructions, and each generated chunk can seed the next through its final frame. We evaluate matched base, prompt-rewritten, and WorldRoll-fine-tuned variants on WorldScore and WorldModelBench. Controlled ablations vary source composition, LoRA rank, data scale, and training duration. Relative to the rewritten-prompt baselines, WorldRoll improves static/dynamic WorldScore by 8.1%/5.5% on LTX and 8.8%/5.9% on CogVideoX-Fun, while raising WorldModelBench by 5.2% and 4.9%, respectively; the gains concentrate in camera control, content alignment, and motion accuracy, with measurable visual-fidelity trade-offs.

Introduction

A useful video world model should continue an observed scene while respecting how the viewpoint, visible geometry, and independently moving subjects are meant to evolve. Existing interactive generators obtain precise control from actions, camera states, or trajectories (Bruce et al. 2024; Valevski et al. 2025; Hu et al. 2023; Zhu et al. 2025; Sun et al. 2025). Camera-conditioned video models likewise show that

explicit geometric inputs can disambiguate physical translation, camera-center rotation, optical zoom, and compound motion (Wang et al. 2024; He et al. 2025; Zheng et al. 2024). Recent systems extend trajectory conditioning to complex I2V paths or redirect the camera of an existing video (Li et al. 2025b; Yu et al. 2025; Bai et al. 2025). WorldRoll instead uses geometric control only to construct supervision and retains an unmodified image-and-language interface at deployment. These interfaces are powerful when calibrated controls are available, but they are less natural for a user who provides one image and says, for example, “move through the doorway, then pan left 60 degrees” (Fig. 1). The user should not need to supply a metric scene scale, a camera matrix, or a frame-aligned trajectory to express such intent.

This paper asks whether precise camera signals can be exploited while supervision is built and then removed from the learned interface. The difficulty is primarily data-centric. Ordinary web-video captions seldom distinguish a truck from a pan, a push-in from optical zoom, or simultaneous motion from sequential motion. Pose-tracked real video offers geometric evidence, but is comparatively narrow in appearance and independent subject dynamics. Camera-controlled synthesis offers exact requested motion, but can inherit teacher artifacts and a restricted motion library. Conversely, diverse real video contains rich scene and subject motion, yet its camera path must be inferred from pixels. No one source provides real appearance, varied dynamics, and reliable camera semantics at once.

We introduce WorldRoll, which combines these signals without altering the deployed I2V interface. WorldRoll constructs three complementary data streams. *Pose-aware real* (PA) video pairs a real continuation with aligned camera poses; a vision-language model (VLM) reads both the video and camera-trajectory statistics to recover a grounded segmented camera path, including rotation angles and clip-local translation magnitudes when supported. *Dynamics-rich real* (DR) video is annotated from pixels alone; the VLM separately describes persistent scene content, independent subject motion, and global camera motion. *Teacher-generated* (TG) video is produced from a real start frame by a camera-controllable world model whose sampled camera path is known by construction. All three streams are mapped to one structured annotation and then rendered deterministically into the same canonical prompt. A second VLM pass forms

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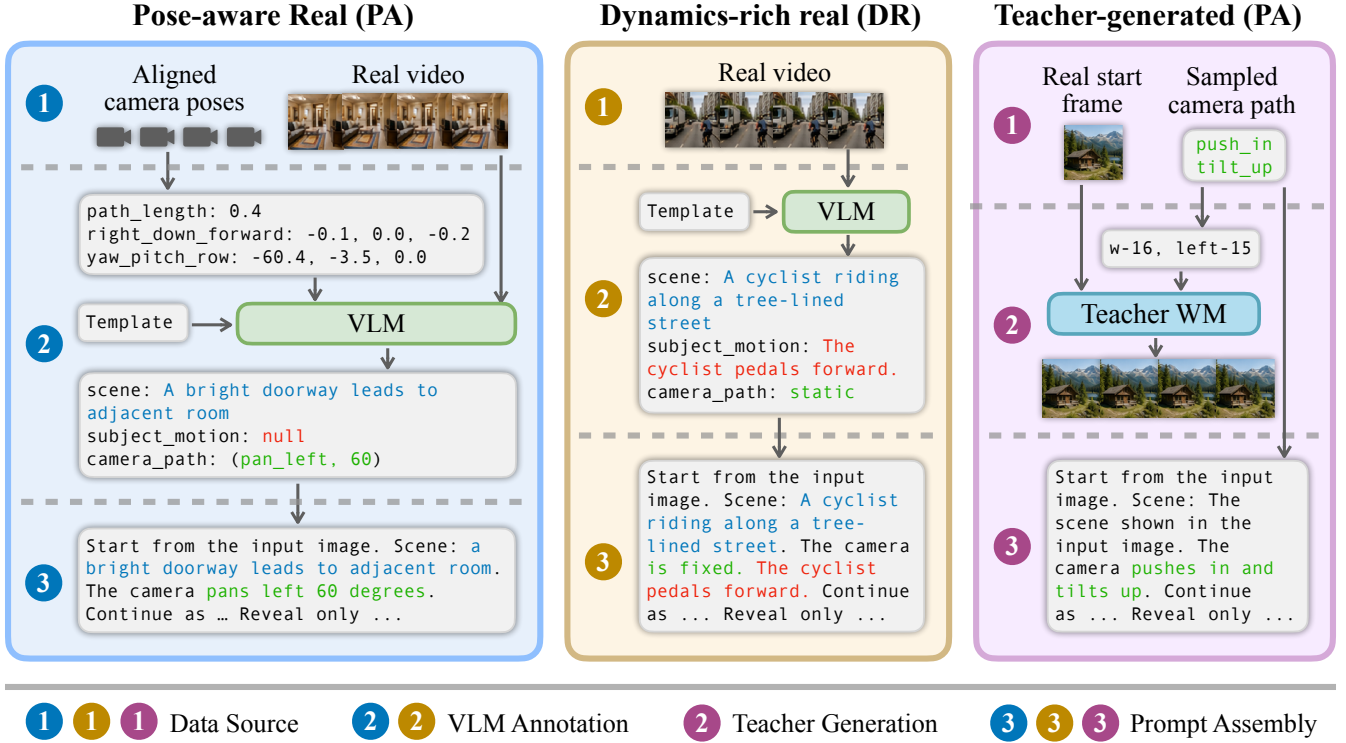


Figure 1: **Data curation for the three WorldRoll streams.** Reading from top to bottom, pose-aware real (PA) curation begins with aligned camera poses and real video, converts the poses into camera-trajectory statistics, and supplies the statistics, video, and a shared annotation template to a VLM; the VLM returns a structured annotation whose fields are deterministically assembled into a canonical prompt. Dynamics-rich real (DR) curation follows the same annotation-and-composition route but infers the structured annotation from real video alone. Teacher-generated (TG) curation pairs a real start frame with a sampled camera path, converts that path into the teacher’s native controls, and uses a camera-controllable teacher to synthesize the target video; the same sampled path is verbalized in the canonical prompt. The illustrated colors trace structured fields into their corresponding prompt spans. Across all streams, privileged poses or teacher controls are retained only for curation and auditing: the student-facing example contains the first frame, canonical prompt, and target video.

a quality gate: it returns structured fatal checks and grouped quality checks, after which code computes a group-balanced quality score and rejects fatal failures or low-scoring candidates (Fig. 2).

The explicit pose or teacher control is privileged information: it helps construct and verify the video–prompt pair but is not an input to the fine-tuned generator. Each student remains an ordinary first-frame-conditioned I2V model, and only LoRA parameters are optimized (Hu et al. 2022) with the backbone’s native diffusion or flow-matching objective (Lipman et al. 2023). At inference, WorldRoll’s prompt rewriter maps arbitrary instructions into the training-time schema. The student then generates a native video chunk from the image and canonical prompt; feeding its final frame into the next call produces a longer rollout, and the instruction may change between chunks. Thus, *pose-free* does not mean camera-control-free: it means that camera intent is expressed in natural language rather than a numeric trajectory or action tensor.

We test whether this transfer principle survives differences in compact I2V design. The two students are LTX-

Video-2B and CogVideoX-Fun-2B (HaCohen et al. 2024; Yang et al. 2025; aigc-apps 2026). Both are approximately 2B-parameter open-weight models, but they use different native frame rates, latent representations, conditioning mechanisms, and learning targets. We evaluate each base checkpoint, its Base_R variant with WorldRoll prompt rewriting, and its WorldRoll-fine-tuned counterpart with the same rewriting on two world-model benchmarks: WorldScore (Duan et al. 2025) and WorldModelBench (Li et al. 2025a). The experiments additionally isolate data composition, adapter capacity, data scale, and training duration.

Our contributions are threefold:

- **Privileged-to-pose-free transfer.** We formulate world-rollout fine-tuning as an offline data-transfer problem: measured or commanded camera trajectories are converted into grounded video-language supervision and discarded before student training.
- **Unified tri-source supervision.** We develop a shared camera language, pose-aware and video-only VLM annotation, camera-controllable teacher generation, and group-balanced quality curation that combine PA, DR,

Stream	Supervision	Primary role
PA	Real video with aligned camera poses	Grounded translation and rotation
DR	Diverse real video annotated from pixels Real start frames rolled out by a camera-controllable teacher	Appearance and independent dynamics
TG		Deliberate camera-path coverage

Table 1: **WorldRoll data streams.** The main training view balances pose-aware real (PA), dynamics-rich real (DR), and teacher-generated (TG) examples under one canonical-prompt I2V contract. The supervision column identifies what is available during curation, not what is exposed to the student.

and TG data under one I2V contract.

- **A controlled two-backbone study.** Across two compact open-weight I2V families and two world-model benchmarks, we separate the effect of prompt rewriting from fine-tuning and measure the effects of source mixture, LoRA rank, data scale, and training duration.

WorldRoll

Problem Formulation and Overview

Let I^0 be an observed image, c a natural-language continuation prompt, and $X = (I^0, I^1, \dots, I^T)$ the corresponding video. WorldRoll fine-tunes a first-frame-conditioned generator to model

$$p_{\theta}(I^{1:T} | I^0, c). \quad (1)$$

During data curation, a sample may additionally have privileged camera evidence Π , such as aligned camera poses or a sampled path consumed by a camera-controllable teacher. WorldRoll uses Π to ground and verify the clip and its language, but the student training record contains only (I^0, c, X) . No numeric pose, trajectory, camera matrix, action tensor, teacher feature, teacher score, or teacher prediction reaches the student.

Figure 1 summarizes how the three data streams convert heterogeneous evidence into the same canonical prompt, and Figure 2 summarizes the common quality gate and pose-free adaptation interface. The streams are deliberately asymmetric: PA contributes grounded camera semantics, DR contributes natural subject and scene dynamics, and TG contributes deliberate path coverage. After curation, however, all accepted examples share the same first-frame, prompt, and target-video contract.

Tri-Source Rollout Data

Table 1 gives a high-level view of the three complementary data streams. PA contributes reliable geometry, DR contributes real-world diversity and independent motion, and TG contributes known camera commands.

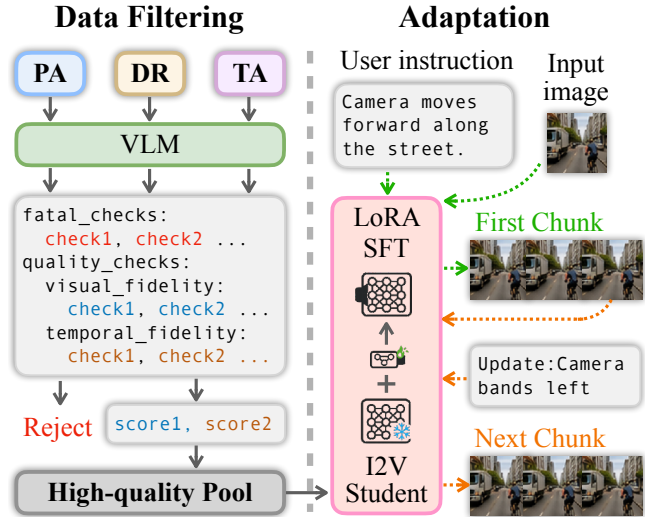


Figure 2: **VLM quality gate and pose-free adaptation.** On the left, candidates from PA, DR, and TG are evaluated using the video, canonical prompt, segmented camera path, and camera-trajectory statistics when available. The VLM returns Boolean fatal checks and grouped quality checks rather than an acceptance decision; code rejects any fatal failure, computes a group-balanced quality score from the applicable quality groups, and rejects candidates below 0.8. On the right, accepted first-frame-prompt-video triples adapt a native I2V student through LoRA fine-tuning. At rollout time, an input image and canonicalized user instruction generate the first chunk; its final frame and an updated instruction condition the next chunk, and the duplicated boundary frame is removed when chunks are concatenated. No pose, path, or structured action is supplied to the student.

Pose-aware real video. PA examples are drawn from RealEstate10K (Zhou et al. 2018). We time-align each five-second clip with its official pose track, invert the provided world-to-camera matrices, and convert the aligned poses into camera-trajectory statistics. The whole-clip statistics include camera-center distance traveled, start-to-end displacement, their ratio, cumulative and endpoint orientation change, and signed translation and yaw/pitch/roll relative to the start camera. The traveled-to-endpoint ratio and cumulative-versus-net rotation help expose reversals or curved paths that an endpoint transform can hide.

Qwen3.6-27B (Qwen Team 2026b) receives sampled video frames, these compact statistics, and the shared annotation template. The video remains authoritative for action meaning and temporal boundaries: pose statistics support direction and magnitude but do not by themselves establish optical zoom, arc, orbit, or a segment boundary. A visually confirmed 57-degree yaw can therefore be rendered as a 60-degree pan, while rotation below 5° over the proposed segment is treated as drift rather than a commanded action. Translations use clip-local units rather than an unsupported metric scale. A magnitude is emitted only when visual motion and camera-trajectory statistics agree; otherwise it is

omitted or the candidate is rejected.

Dynamics-rich real video. DR examples come from OpenVid-1M and Panda-70M (Nan et al. 2025; Chen et al. 2024) after source-level duration, quality, and shot prefiltering. Released web captions and camera tags are not used as WorldRoll annotations. Instead, Qwen3.6 observes the selected clip and independently describes persistent scene content, independent subject motion, and the segmented camera path. This separation is important when, for example, a person walks left while the camera trucks right, or when a pan creates dominant image motion despite a stationary subject. The VLM infers camera motion from coherent global flow, parallax, occlusion changes, and newly revealed regions rather than from local object flow. Because DR has no aligned pose evidence, its camera path carries no numeric magnitude.

Teacher-generated video. TG examples use HY-WorldPlay (Sun et al. 2025) on HunyuanVideo-1.5 (Tencent Hunyuan Foundation Model Team 2025) as an offline camera-controllable teacher. A real start frame is paired with a sampled one- or two-segment camera path, which is converted into the teacher’s native control representation while scene text omits the path. The library contains nine single-segment commands and eight translation-then-pan paths; the two families are sampled with probabilities 0.65 and 0.35, respectively. After generation, the sampled path is verbalized by the same deterministic prompt composer used for real video. The filter still verifies that the generated continuation follows the requested path and preserves the start scene. Thus, the teacher creates selected training videos offline; it provides no logits, features, scores, or denoising trajectories during student optimization.

Unified Camera-Language Representation

Every candidate is normalized to one continuous five-second clip before annotation. Its structured annotation is

$$a = (s, m, \tau), \quad (2)$$

where s is a persistent scene description, m is optional independent subject motion, and τ is a segmented camera path. The path contains $1 \leq J \leq 4$ contiguous segments spanning $[0, 5]$ seconds:

$$\tau = ((t_{j-1}, t_j, \mathcal{A}_j))_{j=1}^J. \quad (3)$$

where $\mathcal{A}_j$ is the set of simultaneous camera actions in segment j ; successive segments are sequential. The representation therefore distinguishes “truck right and pan left” from “truck right, then pan left.”

The closed 19-action ontology is organized into five families: stationary, optical, translation, rotation, and curved. Its actions are *static*; zoom in/out; push/pull, truck left/right, and move up/down; pan left/right, tilt up/down, and roll clockwise/counterclockwise; and arc left/right and orbit left/right. *static* must occur alone, push/pull is not optical zoom, an arc requires curved translation with yaw, and an orbit requires visible motion around a subject or coherent scene region. Paths must cover the full clip without gaps,

overlaps, or zero-length segments. Only PA examples may carry magnitudes: rotations use degrees and translations use clip-local units, while DR and TG paths remain categorical.

The VLM does not directly author the student text. It returns the structured annotation, and a deterministic composer writes the canonical prompt in a fixed order: input-image anchor, persistent scene, segmented camera path, optional independent subject motion, and continuity constraints. For a pose-supported example, the rendered prompt may read:

Start from the input image. Scene: a narrow hallway opens into a furnished room. The camera pushes in 0.4 units, then pans left 60 degrees. A person continues walking through the doorway. Continue as one coherent shot with stable geometry, appearance, identities, and lighting. Reveal only adjacent parts of the same environment; no cuts, jumps, or unrelated elements.

Separating perception from verbalization removes accidental prompt-style variation across streams. Structured annotations and paths remain in the immutable manifest for provenance and analysis, but the student receives only the first frame, canonical prompt, and target video. This composition step is the exact point at which privileged trajectory supervision becomes pose-free language supervision.

Group-Balanced Quality Curation

Annotation and admission are separate VLM passes. After annotation or teacher generation, a second Qwen3.6 pass observes the video, canonical prompt, segmented camera path, and camera-trajectory statistics when available, and returns schema-constrained Booleans. Five fatal checks apply to every stream: a usable video and first frame, a single continuous shot, one coherent world, a fundamentally grounded prompt, and a fundamentally correct camera action, direction, and temporal order. PA additionally requires compatibility with the aligned camera-trajectory statistics and, when present, plausible magnitudes. Uncertainty counts as failure for fatal checks.

Non-fatal quality is grouped by semantic role rather than pooled as one long checklist. The common groups cover visual fidelity, temporal fidelity, world consistency, scene grounding, and camera control. Subject-motion grounding is instantiated only when motion is described; DR adds dynamic-motion quality, TG adds generation fidelity, and PA adds trajectory alignment plus magnitude accuracy when applicable. Action-dependent criteria check, for example, adjacent revealed geometry, plausible parallax, simultaneous versus sequential execution, optical versus physical motion, and arc/orbit geometry only when relevant.

Let $b_{g,m} \in \{0, 1\}$ be criterion m in applicable quality group g for sample x . WorldRoll first averages within each group and then across groups:

$$Q(x) = \frac{1}{|\mathcal{G}(x)|} \sum_{g \in \mathcal{G}(x)} \frac{1}{|\mathcal{C}_g(x)|} \sum_{m \in \mathcal{C}_g(x)} b_{g,m}. \quad (4)$$

A candidate is accepted only if every fatal check passes and $Q(x) \geq 0.8$. The VLM is not allowed to output an acceptance decision or a numeric quality score; both are computed deterministically from its Booleans. Group balancing prevents

Model	S	O	Set.	WS-S	WS-D	Controllability			Quality				Dynamics		
						Cam	Obj	Align	3D	Photo	Style	Qual	Acc	Mag	Sm
Hailuo	–	×	Public	57.55	56.36	22.39	69.56	73.53	67.18	62.82	54.91	52.44	63.46	27.20	70.07
Dynami	1.4B	✓	Public	52.09	47.19	25.15	47.36	25.00	72.90	60.95	78.85	54.40	41.11	39.25	26.92
CogX	5B	✓	Public	62.15	59.12	38.27	40.07	36.73	86.21	88.12	83.22	62.44	69.56	26.42	60.15
Gen-3	–	×	Public	60.71	57.58	29.47	62.92	50.49	68.31	87.09	62.82	63.85	54.53	27.48	68.87
LTX	2B	✓	Base	50.89	51.82	26.40	59.91	42.05	71.02	89.94	41.78	25.15	52.62	28.98	80.31
			Base _R	53.04	53.50	28.73	62.32	54.82	76.49	82.38	41.09	25.45	54.93	32.01	76.74
			Ours	57.33	56.44	31.14	49.33	85.09	71.32	69.81	38.02	56.58	65.86	24.93	72.36
CogX	2B	✓	Base	61.13	58.98	32.38	46.87	51.58	75.49	89.92	70.01	61.68	56.79	26.33	78.76
			Base _R	57.73	57.64	28.47	48.11	60.38	61.34	83.73	63.38	58.71	65.07	25.68	81.49
			Ours	62.82	61.06	31.45	50.31	90.20	71.25	73.61	65.53	57.38	68.87	25.07	76.93

Table 2: **WorldScore**. WS-S and WS-D are the official static- and dynamic-world aggregates; the remaining columns report the benchmark’s controllability, quality, and dynamics dimensions. The upper block gives selected public leaderboard context, while the lower block compares matched Base, Base_R, and Ours rows for both students. Base_R applies the cached WorldRoll rewrite to the base checkpoint, and Ours adds WorldRoll fine-tuning with the same rewrite. S gives the parameter count, O marks open weights, and all percentage scores use the official snapshot (WorldScore Team 2026).

a large visual-fidelity group from outweighing a smaller but control-critical group. Within each stream/source pool, the requested training view selects the highest-quality accepted examples with stable tie-breaking, which preserves exact source proportions. The quality score affects admission and subset construction only; it never weights the student loss.

Pose-Free Student Fine-Tuning and Rollout

Each accepted clip is converted to the native spatial and temporal format of LTX-Video-2B or CogVideoX-Fun-2B. The first frame is the sole visual condition and the canonical prompt is the sole control description. WorldRoll fine-tunes LoRA projections while retaining the pretrained text encoder, latent encoder, scheduler, first-frame-conditioning mechanism, and native diffusion or flow-matching target (Hu et al. 2022; Lipman et al. 2023). Conditioning tokens or conditioned latent regions are excluded from prediction as required by the native backend. No shared auxiliary loss, pose encoder, action adapter, or architecture-specific WorldRoll module is introduced. Consequently, the comparison tests whether one data contract transfers across two compact native model designs rather than whether a new backbone is effective.

At deployment, WorldRoll uses Qwen3.5-2B (Qwen Team 2026a) as a text-only prompt rewriter that maps free-form instructions into the same validated camera-language schema. Because the rewriter cannot inspect the input image, unspecified scene content is represented by a neutral image-grounded phrase rather than invented detail. For a longer rollout, the student generates one native chunk, its final frame initializes the next chunk, and the duplicated boundary frame is removed when chunks are concatenated. The instruction may change at each boundary, allowing a user to compose a longer path through ordinary language while retaining a uniform image-and-text interface.

Experiments

We evaluate whether WorldRoll consistently improves two compact I2V backbones, whether prompt canonicalization explains those gains, and which data and fine-tuning choices are responsible.

Experimental Setup

Models. We study two open-weight first-frame-conditioned students: the LTX-Video 2B v0.9.6 dev checkpoint (HaCohen et al. 2024) and the CogVideoX-Fun V1.1 2B InP checkpoint (Yang et al. 2025; aigc-apps 2026). Both have approximately 2B parameters and generate five-second clips at 512² resolution. LTX uses 121 frames at 24 fps, clean first-frame latent-token conditioning, and a flow-matching target that excludes the conditioned tokens. CogVideoX-Fun uses 41 frames at 8 fps, a first-frame-only inpainting mask, and its native masked-diffusion target.

Training. All variants use LoRA (Hu et al. 2022), bfloat16 precision, AdamW, a constant learning rate of 10^{−5} without decay, and effective batch size 16. Unless an ablation states otherwise, each configuration is trained once, paired configurations share the same fixed training seed, and LoRA alpha equals its rank. The main WorldRoll models use rank 128, 60K training examples with PA:DR:TG = 20:20:20, and the checkpoint at 20K optimizer steps. The Base_R and WorldRoll evaluations use the same cached Qwen3.5-2B rewrites, so prompt processing is not a fine-tuning confound. We retain each backbone’s pretrained text and latent encoders, native objective, scheduler, and first-frame-conditioning mechanism.

Benchmarks. WorldScore (Duan et al. 2025) measures controllability, quality, and dynamics through ten fine-grained dimensions and static- and dynamic-world aggregates. WorldModelBench (Li et al. 2025a) scores instruction following, physical adherence, and common sense. These two world-model benchmarks form the main evaluation and are reported separately in Tables 2 and 3.

Model	S	O	Set.	WMB	IF	Phys	CS
Kling	–	×	Public	9.10	2.32	4.81	1.95
MiniMax	–	×	Public	8.92	2.28	4.72	1.92
CogX	5B	✓	Public	7.08	1.78	4.17	1.12
OpenSora	–	✓	Public	5.82	1.42	3.85	0.55
LTX	2B	✓	Base	7.35	1.64	4.49	1.22
			Base _R	7.44	1.68	4.51	1.25
			Ours	7.83	1.85	4.57	1.41
CogX	2B	✓	Base	7.58	1.89	4.37	1.31
			Base _R	7.73	1.95	4.44	1.34
			Ours	8.11	2.20	4.50	1.41

Table 3: **WorldModelBench.** WMB sums instruction following (IF; 0–3), physical adherence (Phys; 0–5), and common sense (CS; 0–2) to a maximum of 10; totals are computed before the displayed components are rounded. The upper block gives selected public leaderboard context, while the lower block compares matched Base, Base_R, and Ours rows for both students. Base_R applies the cached WorldRoll rewrite to the base checkpoint, and Ours adds WorldRoll fine-tuning with the same rewrite. S gives the parameter count, O marks open weights, and public values use the official snapshot (WorldModelBench Team 2026).

Main Results

Tables 2 and 3 contain three rows per student: the base checkpoint on the original benchmark prompt, the same base checkpoint with WorldRoll prompt rewriting (Base_R), and the WorldRoll-fine-tuned checkpoint with that identical rewrite. Within each local triplet, the input image, sampler, seed, and all non-model settings are matched. Each benchmark case is generated once per local row. Public leaderboard rows provide benchmark-specific context rather than matched reproductions because their generation protocols can differ. Every local WorldRoll row uses the rank-128 checkpoint trained on the 60K 20:20:20 data view for 20K steps.

WorldScore. Table 2 separates the benchmark-level scores from the fine-grained controllability, quality, and dynamics metrics. Prompt rewriting alone is backbone-dependent: it raises LTX’s WS-S/WS-D by 2.15/1.68 points but lowers CogX’s by 3.40/1.34 points. WorldRoll fine-tuning more than recovers this loss and consistently improves over the matched rewritten baseline, increasing WS-S/WS-D by 4.29/2.94 points for LTX and 5.09/3.42 points for CogX. The largest shared improvements occur in content alignment (+30.27 and +29.82) and motion accuracy (+10.93 and +3.80), while camera control also rises for both students (+2.41 and +2.98). These control gains are not free: photometric consistency drops by 12.57/10.12 points, motion smoothness by 4.38/4.56 points, and motion magnitude by 7.08/0.61 points relative to Base_R. The opposing trends suggest that the current training distribution prioritizes instruction execution and stronger motion over conservative reconstruction of the pre-trained appearance and temporal priors.

WorldModelBench. WorldModelBench asks whether a generated continuation follows the instruction and remains

Data (K)			WMB	WS-S*	WS-D*
PA	DR	TG			
0	0	10	7.51	52.57	52.11
5	0	5	7.56	52.31	51.82
5	5	0	7.57	52.55	52.84
0	5	5	7.63	52.02	53.25
2	3	5	7.59	52.34	53.34
3	3	4	7.60	52.62	53.20

Table 4: **Data-composition ablation.** LTX-Video-2B is fine-tuned for 10K updates with rank/alpha 32/32 on each 10K mixture of pose-aware real (PA), dynamics-rich real (DR), and teacher-generated (TG) examples. WMB uses the full benchmark; WS-S* and WS-D* are the static- and dynamic-world aggregates on the same fixed, task-stratified 360-example WorldScore subset. The starred scores support only within-table comparison and are not comparable to full WorldScore results.

Model	LoRA rank r				
	16	32	64	128	256
LTX	7.51	7.53	7.55	7.69	7.62
CogX	7.94	7.93	7.99	8.01	8.00

Table 5: **LoRA capacity.** Cells report WMB for single LTX-Video-2B and CogVideoX-Fun-2B runs after 10K updates on the same 10K 3:3:4 data mixture, varying rank/alpha together from 16 to 256. All configurations use effective batch size 16 and a constant 10^{-5} learning rate.

Data (K)			WMB at step (K)					
PA	DR	TG	5	10	15	20	25	30
6	6	8	7.58	7.65	7.70	7.65	7.71	7.70
12	12	16	7.65	7.70	7.74	7.71	7.72	7.68
20	20	20	7.71	7.74	7.77	7.83	7.74	7.71

Table 6: **Data and training scale.** LTX-Video-2B uses rank/alpha 128/128, effective batch size 16, and a constant 10^{-5} learning rate on nested 20K, 40K, and 60K PA/DR/TG views. The left block gives each source composition; the right block reports WMB every 5K updates through 30K, separating matched-checkpoint data-scale effects from within-row optimization duration.

physically and semantically plausible. Here prompt rewriting is uniformly beneficial: WMB rises by 0.09 points for LTX and 0.15 points for CogX, with gains in every component. Fine-tuning then adds 0.39 and 0.38 points, respectively, for a mean improvement of 0.39 points over Base_R. The gains again span every component: instruction following improves by 0.17/0.25 points, physical adherence by 0.06/0.06 points, and common sense by 0.16/0.07 points for LTX/CogX. Thus, the learned improvement is not explained by canonical phrasing alone and extends beyond camera-language matching to the physical and semantic plausibility of the continuation.

Figure 3 complements the benchmark results with repre-

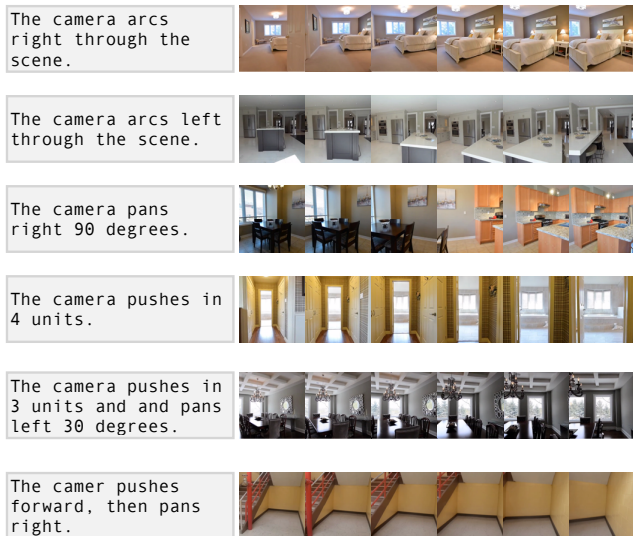


Figure 3: **Precise camera control with WorldRoll.** Representative rollouts demonstrate bidirectional arcs, specified pan and push magnitudes, simultaneous translation and rotation, and sequential motion. WorldRoll receives only the first frame and text; frames are sampled uniformly in time.

representative WorldRoll outputs. The camera follows directional arcs, magnitude-conditioned pans and pushes, and simultaneous or ordered actions while revealing coherent scene continuations. The opposing arcs test direction rather than generic lateral drift, while the magnitude-conditioned and compound cases distinguish rotation from translation and simultaneous from ordered execution. These examples demonstrate language-only execution without exposing the privileged trajectories used during data construction.

Data Composition

We isolate the contribution of each source type with LTX-Video-2B, 10K-example training views, LoRA rank/alpha 32/32, effective batch size 16, a constant 10^{-5} learning rate, and 10K updates. Table 4 compares one teacher-only endpoint, three pairwise mixtures, and two mixtures containing all three source types. Each run is evaluated on WMB and a fixed, task-stratified subset of 360 WorldScore examples, reporting WS-S* and WS-D*. The subset reduces evaluation cost and provides a complementary view, but its scores are comparable only within this ablation. No mixture wins all three metrics: 0:5:5 gives the highest WMB, 2:3:5 gives the highest WS-D*, and 3:3:4 gives the highest WS-S*. The selected 3:3:4 view is within 0.03 WMB and 0.14 WS-D* of the respective maxima while retaining PA’s grounded numeric camera magnitudes. We use it for the capacity study and as the starting composition for scaling; the 60K view contains 20K examples from each stream.

Adapter Capacity

With the 10K 3:3:4 mixture fixed, we train both students at LoRA ranks 16, 32, 64, 128, and 256 for 10K updates with

effective batch size 16 and a constant 10^{-5} learning rate. Table 5 reports WMB. This ablation determines whether performance is monotonic in adapter rank or saturates at a smaller capacity. LTX improves from 7.51 at rank 16 to a peak of 7.69 at rank 128, then drops to 7.62 at rank 256. CogX is nearly flat at low rank, likewise peaks at 8.01 at rank 128, and differs by only -0.01 at rank 256. Rank 128 is therefore the shared optimum for both students; we use rank/alpha 128/128 for all larger-scale and main runs.

Data and Training Scale

Finally, we train LTX-Video-2B with rank 128 on three nested data views: 6+6+8 (20K), 12+12+16 (40K), and 20+20+20 (60K), where the terms denote PA, DR, and TG counts in thousands. Every run uses effective batch size 16 and a constant 10^{-5} learning rate, continues to 30K optimizer steps, and is evaluated every 5K steps in Table 6. This experiment tests the separate effects of unique-data scale and optimization duration at matched checkpoints. At every checkpoint through 25K steps, increasing the data view improves WMB; at the selected 20K checkpoint, the score rises monotonically from 7.65 to 7.71 to 7.83 as the view grows from 20K to 60K examples. Longer optimization is not uniformly beneficial under the constant learning rate and small global batch: the 40K and 60K runs peak at 15K and 20K steps, respectively, and all three runs finish below their best observed score at 30K. We therefore use the 60K, 20K-step checkpoint for both main students, which is the overall maximum in the sweep rather than the final checkpoint.

Conclusion

WorldRoll converts privileged camera trajectories into pose-free language supervision. Aligned poses and sampled teacher paths guide data construction and verification but are discarded before fine-tuning two native I2V students. Against the same rewritten-prompt baselines, WorldRoll raises WS-S/WS-D by 4.29/2.94 points for LTX and 5.09/3.42 points for CogVideoX-Fun, and WMB by 0.39 and 0.38 points. The strongest shared gains occur in camera control, content alignment, and motion accuracy, while fidelity regressions expose a remaining trade-off. Ablations support a PA-preserving 3:3:4 compromise and rank 128; WMB increases through the 60K view at the 20K checkpoint, where the corresponding run also peaks.

Because both students retain their native objectives and conditioning mechanisms, the gains across backbones support the shared data contract rather than an architecture-specific control module. The simultaneous fidelity losses, however, show that stronger instruction execution should not be interpreted as uniformly better video generation.

Pose-free language control cannot guarantee exact metric tracking, autoregressive rollouts can drift, and the single-run results do not estimate training-seed variability. Future work can add richer paths, closed-loop correction, broader interactive actions, and fidelity-preserving objectives. Subject to upstream licenses, we plan to release construction code, prompts and manifests, evaluation recipes, and LoRA weights.

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